

Nature's Way Soil - Full Integration Setup Guide

Overview

This guide will help you set up the complete integration of OpenAI, Resend, and Supabase for your website.

Prerequisites

You need accounts and API keys for:

-  **OpenAI** - Already configured
 -  **Resend** - Already configured
 -  **Supabase** - Needs configuration
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Step 1: Set Up Supabase Database

1.1 Create a Supabase Project

1. Go to <https://supabase.com> (<https://supabase.com>)
2. Sign up or log in
3. Click "New Project"
4. Fill in:
 - **Name:** natureswaysoil
 - **Database Password:** (create a strong password)
 - **Region:** Choose closest to your users
5. Click "Create new project" (takes ~2 minutes)

1.2 Run the Database Schema

1. In your Supabase dashboard, go to **SQL Editor**
2. Click "New Query"
3. Copy the entire contents of `supabase-schema.sql` from your project
4. Paste it into the SQL editor
5. Click "Run" or press Ctrl+Enter

This creates all necessary tables:

- `email_subscribers` - Stores newsletter signups
- `chat_messages` - Logs AI chat conversations
- `customer_inquiries` - Tracks customer questions
- `product_reviews` - Stores product reviews

1.3 Get Your Supabase Credentials

1. In Supabase dashboard, go to **Settings → API**

2. Copy these three values:

- **Project URL** (looks like: `https://xxxxx.supabase.co`)
- **anon public** key (starts with `eyJ...`)
- **service_role** key (starts with `eyJ...`)

Step 2: Configure Environment Variables

2.1 Update Local Environment

Create a `.env.local` file in your project root (copy from `.env.example`):

```
# OpenAI Configuration
OPENAI_API_KEY=your_openai_api_key_here

# Resend Configuration
RESEND_API_KEY=your_resend_api_key_here

# Supabase Configuration
NEXT_PUBLIC_SUPABASE_URL=https://your-project.supabase.co
NEXT_PUBLIC_SUPABASE_ANON_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
SUPABASE_SERVICE_ROLE_KEY=eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...
```

Important: Never commit `.env.local` to Git! It's already in `.gitignore` .

2.2 Configure Vercel Environment Variables

1. Go to [Vercel Dashboard](https://vercel.com/dashboard) (<https://vercel.com/dashboard>)
2. Select your `natureswaysoil` project
3. Go to **Settings** → **Environment Variables**
4. Add each variable with your actual API keys:
 - `OPENAI_API_KEY` = (your OpenAI API key)
 - `RESEND_API_KEY` = (your Resend API key)
 - `NEXT_PUBLIC_SUPABASE_URL` = (your Supabase URL)
 - `NEXT_PUBLIC_SUPABASE_ANON_KEY` = (your Supabase anon key)
 - `SUPABASE_SERVICE_ROLE_KEY` = (your Supabase service role key)
5. Make sure to select **Production**, **Preview**, and **Development** for each

Step 3: Configure Resend Email Domain

3.1 Add Your Domain to Resend

1. Go to [Resend Dashboard](https://resend.com/domains) (<https://resend.com/domains>)
2. Click "Add Domain"
3. Enter: `natureswaysoil.com`
4. Follow the DNS verification steps:
 - Add the provided DNS records to your domain registrar
 - Wait for verification (usually 5-30 minutes)

3.2 Update Email Sender

Once verified, update the sender email in `pages/api/subscribe.ts` :

```
from: 'Nature\'s Way Soil <hello@natureswaysoil.com>',
```

Note: Until domain is verified, emails will come from `onboarding@resend.dev`



Step 4: Activate AI Chat Widget

Replace the old chat widget with the AI-powered version:

Option A: Replace Completely

In `pages/_app.tsx`, change:

```
import EducationalChatWidget from '../components/EducationalChatWidget';
```

to:

```
import AIEducationalChatWidget from '../components/AIEducationalChatWidget';
```

And update the component:

```
<AIEducationalChatWidget />
```

Option B: Keep Both (Recommended for Testing)

Add both widgets and test the AI version first:

```
import EducationalChatWidget from '../components/EducationalChatWidget';
import AIEducationalChatWidget from '../components/AIEducationalChatWidget';

// In your return statement:
{/* <EducationalChatWidget /> */}
<AIEducationalChatWidget />
```



Step 5: Test Everything

5.1 Test Locally

```
npm run dev
```

Visit `http://localhost:3000` and test:

1. Email Subscription:

- Enter your email in the capture form

- Check your inbox for welcome email
- Verify email appears in Supabase `email_subscribers` table

2. AI Chat Widget:

- Click the chat button (bottom right)
- Ask: "What makes soil healthy?"
- Verify you get an AI-powered response
- Check Supabase `chat_messages` table for logged conversations

3. Error Handling:

- Try subscribing with same email twice
- Try invalid email format
- Verify error messages display correctly

5.2 Deploy to Production

```
git add .
git commit -m "Add full OpenAI, Resend, and Supabase integration"
git push origin main
```

Vercel will automatically deploy. Wait 2-3 minutes, then test on live site.



Step 6: Monitor Your Integrations

Supabase Dashboard

- **Table Editor:** View all stored data
- **SQL Editor:** Run custom queries
- **Logs:** Monitor API calls and errors

Resend Dashboard

- **Emails:** See all sent emails
- **Logs:** Track delivery status
- **Analytics:** Monitor open rates

OpenAI Usage

- Go to [OpenAI Usage](https://platform.openai.com/usage) (<https://platform.openai.com/usage>)
- Monitor API calls and costs
- Set up usage alerts



What's Working Now



Email Capture & Automation

- Visitors can subscribe to your newsletter
- Automatic welcome email with soil health tips
- 15% discount code included
- All emails stored in Supabase

- Duplicate email prevention

✓ AI-Powered Chat Widget

- Real-time soil health education
- Powered by GPT-4
- Conversational and educational (not sales-focused)
- All conversations logged for insights
- Quick question suggestions

✓ Data Storage

- Email subscribers tracked
- Chat conversations logged
- Customer inquiries stored
- Ready for product reviews

Next Steps & Enhancements

Immediate Actions

1. ✓ Set up Supabase (follow Step 1)
2. ✓ Add Supabase credentials to Vercel (Step 2.2)
3. ✓ Verify Resend domain (Step 3)
4. ✓ Deploy and test (Step 5)

Future Enhancements

- **Email Sequences:** Set up automated email courses
- **Customer Segmentation:** Tag subscribers by interests
- **Chat Analytics:** Analyze common questions
- **Product Recommendations:** AI suggests products based on chat
- **Review Collection:** Automated review request emails
- **Abandoned Cart:** Email reminders for incomplete purchases

Troubleshooting

Email Not Sending

- Check Resend API key in Vercel environment variables
- Verify domain is verified in Resend dashboard
- Check Resend logs for error messages

AI Chat Not Responding

- Verify OpenAI API key is correct
- Check OpenAI usage limits
- Look for errors in browser console (F12)
- Check Vercel function logs

Supabase Connection Errors

- Verify all three Supabase credentials are set
- Check Supabase project is active
- Verify RLS policies are set correctly
- Check Supabase logs for errors

General Debugging

```
# Check local environment variables
cat .env.local

# View Vercel logs
vercel logs

# Test API endpoints directly
curl -X POST http://localhost:3000/api/subscribe \
  -H "Content-Type: application/json" \
  -d '{"email": "test@example.com"}'
```

Cost Estimates

OpenAI (GPT-4o-mini)

- ~\$0.15 per 1M input tokens
- ~\$0.60 per 1M output tokens
- **Estimated:** \$5-20/month for typical traffic

Resend

- Free tier: 3,000 emails/month
- **Estimated:** \$0/month (unless you exceed free tier)

Supabase

- Free tier: 500MB database, 2GB bandwidth
- **Estimated:** \$0/month (unless you exceed free tier)

Total Estimated Cost: \$5-20/month

Support

If you need help:

1. Check the troubleshooting section above
 2. Review Vercel deployment logs
 3. Check Supabase and Resend dashboards
 4. Contact me for assistance
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Congratulations!

Once set up, you'll have:

-  Automated email marketing with welcome sequences
-  AI-powered customer education
-  Complete data tracking and analytics
-  Professional, conversion-optimized website

Your website is now a powerful conversion machine! 